MILITARY SPECIFICATION

RADAR ALTIMETER WARNING SET AN/APQ-107 (*)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide an aural warning signal to energize the pilot's and copilot's headsets and a visual warning signal to illuminate a visual warning feature in accordance with the operational requirements specified herein.

1.2 Classification - The Radar Altimeter Warning Set shall consist of a single unit designated AN/APQ-107.

*1.3 Associated Equipment - The equipment shall operate satisfactorily when either a single installation or two separate installations are made in the same aircraft, and shall operate with the following associated equipment which shall not be supplied as part of this equipment:

(1) Radar Set AN/APN-117 (Radar Altimeter), Radar Set AN/APN-22 (Radar Altimeter) or Electronic Altimeter Set AN/APN-141 (V).

(2) Intercommunication Set AN/AIC-4A (with Interphone Amplifier CW1003C only), -14, -14A, -15, and -22.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of

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MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specifications for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-E-19600	Electronic Modules, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25237	Lamp, Incandescent, T-1-3/4 Bulb, Midget Flange Base

DRAWINGS

Bureau of Naval Weapons

65A27F4	Installation Drawing, Warning Set, Radar Altimeter, AN/APQ-107
64A54A100	Switch, Pressure, Differential, Altitude
65A27D11	Synchro Data Converter
65A27D21	Power Supply

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65A27D101	Differential Trigger Assembly A1
65A27D201	Wheel Logic Assembly A2
65A27D301	Warning Logic Assembly A3
65A27D401	Delayed Reliability Logic Assembly A4
65A27D501	Gate Generator Assembly A5
65A27D601	Light Driver and Osc Assembly A6
65A27D701	1 KC Amplifier Assembly A7 or A8
65A27D1	Warning Set, Radar Altimeter, AN/APQ-107
65A27F2	Electrical Schematic, Warning Set, Radar Altimeter, AN/APQ-107

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 201, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and Materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400 shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.4). Electronic modules shall be in accordance with Specification MIL-E-19600.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

* 3.1.2 Electron Devices - Solid state design shall be employed throughout. Transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.2 Design and Construction - The equipment shall conform with all applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

* 3.2.1 Total Weight - The total weight of the equipment shall be a minimum consistent with good design and shall not exceed nine pounds.

3.2.2 Reliability -

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity.

* 3.2.2.3 Reliability in Mean-Time-Between-Failures (MTBF) - The equipment shall have 1100 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

* 3.2.4 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

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3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

*3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1 V AC and 27.5 ± 0.5 V DC

3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

*3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

*3.2.9.2 Temperature - The minimum operating temperature shall be -44°C in lieu of -55°C.

*3.2.9.3 Altitude - The maximum operating altitude shall be 30,000 feet in lieu of 70,000 feet.

3.2.10 Warm-Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed one minute under standard conditions and three minutes at extreme service conditions.

*3.2.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

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(1) AC Power (Single Phase) 115V, Category "B", 50 VA

(2) DC Power 28 V, Category "B", 1.2 Amps

* 3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions.

* 3.3.1 Operation - The Radar Altimeter Warning Set, when connected in accordance with BuWeps Installation Control Drawing 65A27F4, shall provide the following functions in accordance with the detailed requirements of paragraph 3.4:

(1) Provide separate aural warning output signals to the pilot's and copilot's headsets and a visual warning output signal to the visual warning feature at two discrete altitude settings. The altitude settings at which the warning signal shall be initiated are designated herein as the high altitude index and the low altitude index.

(2) Provide signals identical to those of paragraph 3.3.1(1) when the aircraft is at or below the pilot's and/or copilot's radar altimeter limit indicator setting.

(3) Provide signals identical to those of paragraph 3.3.1(1) when there is a power failure to the associated radar altimeter.

(4) Provide signals identical to those of paragraph 3.3.1(1) indicating when the associated radar altimeter is operating in an unreliable mode.

(5) Provide signals identical to those of paragraph 3.3.1(1) in response to a local or remote test command. The test command shall simulate "gear-up" flight at an altitude below ten feet and be available at any time.

(6) Provide a delayed reliability signal to the Automatic Stabilization Equipment (ASE).

3.4 Detailed Requirements -

3.4.1 High Altitude Index - The high altitude index shall be adjustable on the ground between the altitudes of 100 and 500 feet. The settings of the altitude index shall be maintained to an accuracy of ± 5%.

3.4.2 Low Altitude Index -

* 3.4.2.1 For the AN/APN-117 Radar Altimeter the low altitude index shall be adjustable on the ground between the altitudes of 8 and 300 feet. The setting of the altitude index shall be maintained to an accuracy of ± 5 feet for settings below 100 feet, and ± 5% for settings above 100 feet.

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*3.4.2.2 For the AN/APN-22 and AN/APN-141 Radar Altimeters the low altitude index shall be adjustable on the ground between the altitudes of 25 and 300 feet. The setting of the altitude index shall be maintained to an accuracy of ± 5 feet for settings below 100 feet and ± 5% for settings above 100 feet.

*3.4.3 Description of Output Signals -

3.4.3.1 Electrical Aural Warning Signal Characteristics - The output signal when enabled as described herein shall be a 1000 ± 200 cps sine wave voltage which is pulse modulated on and off at a rate of 2 ± 0.5 pulses per second with a pulse width of 0.2 ± 0.05 seconds. The voltage amplitude shall be ground adjustable through a range of 5.6 to 30 volts peak-to-peak and shall maintain the set level to an accuracy of ± 20%. This level shall be attained when the output is terminated in an external resistance of 1200 ohms. The maximum residual output during the "OFF" portion of the cycle shall be less than 0.03 volts peak-to-peak across the external load. During the "ON" portion of the cycle the harmonic distortion shall be less than 40% and the amplitude modulation shall be less than 10%. The output signal gate shall be delayed 0.75 ± 25% seconds after application of the enabling signal of paragraphs 3.4.3.4.1, 3.4.3.4.2, 3.4.3.4.3, 3.4.3.4.5, or 3.4.3.4.6.

3.4.3.2 Electrical Visual Warning Signal Characteristics - The output signal when enabled as described herein shall consist of pulses of DC voltage which shall be gated on and off synchronously with the signal of paragraph 3.4.3.1. When in the "ON" condition, the pulse amplitude shall be 26 ± 2 V DC for a minimum of 75% of the pulse duration when the output is terminated with an external visual warning feature consisting of three parallel connected MS 25237-327 incandescent lamps. When not gated "ON", the output signal shall be less than 0.5 volts. The output signal gate shall be delayed 0.75 ± 25% seconds after application of the enabling signal of paragraphs 3.4.3.4.1, 3.4.3.4.2, 3.4.3.4.3, 3.4.3.4.5, or 3.4.3.4.6.

3.4.3.3 Delayed Reliability Signal Characteristics - The signal shall be + 27.0 ± 2 V DC when the output voltage is terminated with an external resistance of 270 ohms. This output voltage shall be less than + 1 V DC when the reliability input signal, as defined herein, is in the unreliable state and shall be switched to + 27.0 ± 2 V DC after a variable delay of 3 to 9 seconds upon restoration of the reliability voltage. The setting of the variable delay shall be maintained within ± 25%.

3.4.3.4 Provisions for Enabling Output Signals -

3.4.3.4.1 High Altitude - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled for a period of 3 ± 1 seconds when the altitude input signal as defined herein has decreased to the high altitude index setting. The output shall not be enabled when the altitude input signal is increased to the high altitude index setting from below this setting.

3.4.3.4.2 Low Altitude - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled and remain enabled while the altitude input signal is at or below the low altitude index setting.

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3.4.3.4.3 Radar Altimeter Limit Indicator - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled and remain enabled when the pilot's and/or copilot's radar altimeter limit indicator input signal as defined herein is in the "ON" condition.

3.4.3.4.4 Altimeter Power Failure - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled and remain enabled during the time when the altimeter indicator power failure signal as herein defined is in the "OFF" condition.

3.4.3.4.5 Altimeter Unreliable - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled and remain enabled when the reliability input signal is in the unreliable state.

3.4.3.4.6 Test Command - The output signals of paragraphs 3.4.3.1 and 3.4.3.2 shall be enabled and remain enabled during the application of the test command signal as defined herein.

3.4.3.5 Provisions for Disabling Output Signals - The disabling functions herein defined shall be designed such that loss of any disabling signal shall not result in a failure to obtain the output signals under the conditions of paragraph 3.4.3.4.

3.4.3.5.1 Landing Gear Inhibit #1 - The output signals under the conditions of paragraph 3.4.3.4.5 shall be disabled when all the wheel down and locked input signals as defined herein are in the down and locked states.

3.4.3.5.2 Landing Gear Inhibit #2 - The output signal of paragraph 3.4.3.1 shall be disabled under the conditions of paragraphs 3.4.3.4.2 and 3.4.3.4.3 when all the wheel down and locked input signal voltages are in the down and locked states.

3.4.3.5.3 Landing Gear Inhibit #3 - The output signals under the conditions of paragraph 3.4.3.4.1 shall be disabled when all the wheel down and locked input signals are in the down and locked states and the Wheel Inhibit Signal and High Altitude Inhibit Input are connected in accordance with BuWeps Installation Control Drawing 65A27F4.

3.4.3.5.4 Automatic Stabilization Equipment - The output signals under the condition of paragraph 3.4.3.4.1 shall be disabled when the ASE engagement input signal is in the "ON" condition.

3.4.3.5.5 Pressure Altitude - The output signal under the condition of paragraphs 3.4.3.4.1, 3.4.3.4.2, 3.4.3.4.3, and 3.4.3.4.5 shall be disabled when the pressure altitude input signal as defined herein is actuated and the delayed reliability output signal is less than + 1 V DC.

3.4.3.5.6 Weight-On-Wheel - The output signal of paragraph 3.4.3.1 shall be disabled under the conditions of paragraph 3.4.3.4.4 and when the weight-on-wheel input signal as defined herein is in the "ON" condition.

3.4.4 Description of Input Signals -

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3.4.4.1 Altitude Input Signals -

3.4.4.1.1 Altimeter Set, Electronic AN/APN-141(V) - The altitude input signal voltage from this altimeter is a DC voltage (V) defined by the following equations:

$$V = E_1A + C_1, \quad 0 \leq A \leq 400 \text{ feet}$$

$$V = E_2A + C_2, \quad 400 \leq A \leq 500 \text{ feet}$$

where $E_1 = 2.0 \pm 0.005$ volts per 100 feet

$E_2 = 2.61 \pm 0.0065$ volts per 1000 feet

$C_1 = 1 \pm 0.0025$ volts

$C_2 = 7.95 \pm .02$ volts

The input impedance presented to this signal shall be greater than 100,000 ohms.

3.4.4.1.2 Radar Set AN/APN-117 (Radar Altimeter) - The altitude input signals from this altimeter are 400 cps stator voltages from a synchro torque transmitter and are as defined by the following equations:

$$E(S1\ S3) = N\ E(R2\ R1) \sin -a$$

$$E(S3\ S2) = N\ E(R2\ R1) \sin (-a + 120^\circ)$$

$$E(S2\ S1) = N\ E(R2\ R1) \sin (-a + 240^\circ)$$

where: The terminal sequence as indicated in the parentheses designates the sense of the voltage vector.

N is $0.783 \pm 2\%$

E(R2 R1) is 115 V RMS, 400 cps

a is the shaft angle displaced in a positive direction from synchro zero.

and: The relationship between altitude and shaft angle is described by the following:

(1) $a = 1.988A, \quad 0 \leq A \leq 100$ feet

(2) $a = .512\ A + 147.6, \quad 100 \leq A \leq 200$ feet

(3) $a = 231 + 10\ (\frac{A}{100}) - (\frac{A}{200})^2, \quad 200 \leq A \leq 500$ feet

The input impedance presented to each of the stator voltages shall be greater than 2,000 ohms line-to-line.

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*3.4.4.1.3 Radar Set AN/APN-22 (Radar Altimeter) - The altitude input signals from this altimeter are 400 cps stator voltages from a synchro torque transmitter and are as defined by the following equations:

$$E(S1\ S3) = N\ E(R2\ R1)\ \sin(-a)$$

$$E(S3\ S2) = N\ E(R2\ R1)\ \sin(-a + 120^{\circ})$$

$$E(S2\ S1) = N\ E(R2\ R1)\ \sin(-a + 240^{\circ})$$

where: The terminal sequence as indicated in the parentheses designates the sense of the voltage vector.

N is $0.783 \pm 2\%$

E(R2 R1) is 115 V RMS, 400 cps

a is the shaft angle displaced in a positive direction from synchro zero.

and: The relationship between altitude and shaft angle is described by the following:

(1) $a = 0.60A \qquad 0 \leq A \leq 200$ feet

(2) $a = 101 + 10\left(\frac{A}{100}\right) - \left(\frac{A}{200}\right)^2, \quad 200 \leq A \leq 500$ feet

The input impedance presented to each of the stator voltages shall be greater than 2,000 ohms line-to-line.

3.4.4.2 Reliability Input Signal - The reliability input signal is a DC voltage from the Radar Altimeter between +3 and +29 volts when the altimeter is reliable and less than one volt when the altimeter is unreliable. The input impedance presented to this signal shall be greater than 5,000 ohms.

3.4.4.3 ASE Engagement Input Signal - The ASE engagement input signal is a DC voltage of $+25 \pm 4$ volts when the ASE coupler is engaged ("ON" condition) and zero volts when the ASE coupler is disengaged ("OFF" condition). The input impedance presented to this signal shall be greater than 10,000 ohms.

3.4.4.4 Altimeter Indicator Power Failure Input Signal - The altimeter power failure input signal voltage is 105 - 122 V RMS, 380 - 420 cps in the "ON" condition and zero volts in the "OFF" condition. The input impedance presented to this signal shall be greater than 30,000 ohms.

3.4.4.5 Wheel Down and Locked Input Signals - The wheel down and locked input signal voltage is aircraft DC ground when the wheel is down and locked and is between + 4.5 and + 29 volts when the wheel is not down and locked. The input impedance presented to each signal shall be greater than 10,000 ohms.

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3.4.4.6 Test Command Input Signal - The test command input signal will be +25 ± 4 V DC when the "press to test" switch is depressed and open circuited at all other times. The input impedance presented to this signal shall be greater than 200 ohms.

3.4.4.7 Pressure Altitude Input Signal - The equipment shall contain a barometric switch. The switch shall be closed at a pressure differential corresponding to an altitude increase of 700 ± 200 feet with respect to the take-off altitude and shall be open at any altitude below 700 ± 200 feet with respect to the take-off altitude. The maximum take-off altitude will be limited to 6,000 feet for this consideration. The reference altitude shall be established upon application of aircraft DC power.

3.4.4.8 Radar Altimeter Limit Indicator Input Signal - The radar altimeter limit indicator input signal is 7 to 28 V AC or +17 to +29 V DC when the altitude is at or below the setting of the altimeter limit indicator ("ON" condition) and less than one volt when the altitude is greater than the setting of the altimeter limit indicator ("OFF" condition). The input impedance presented to this signal shall be greater than 2,000 ohms.

3.4.4.9 Weight-On-Wheel Input Signal - The weight-on-wheel input signal is aircraft DC ground in an "ON" condition and either aircraft +28 V DC or open circuited in an "OFF" condition. The input impedance presented to this signal shall be greater than 20,000 ohms.

*3.4.5 Subassembly Electrical Connections - Modular breakdown and internal wiring connections shall conform to the requirements of BuWeps Electrical Schematic 65A27F2.

*3.4.6 Internal Mechanical Configuration - The internal configuration shall conform to the requirements of BuWeps Specification Control Drawing 65A27D1.

*3.4.7 Contents - The Warning Set shall contain the following subassemblies. Subassemblies shall be in accordance with the requirements of the respective BuWeps Specification Control Drawing.

SUBASSEMBLY	QUANTITY/UNIT	SPECIFICATION CONTROL DRAWINGS
(1) Barometric Switch	1	64A54A100-2
(2) Synchro Data Converter	1	65A27D11
(3) Power Supply	1	65A27D21
(4) Differential Trigger Module	1	65A27D101
(5) Wheel Logic Module	1	65A27D201
(6) Warning Logic Module	1	65A27D301
(7) Delayed Reliability Logic Module	1	65A27D401

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SUBASSEMBLY	QUANTITY/UNIT	SPECIFICATION CONTROL DRAWINGS
(8) Gate Generator Module	1	65A27D501
(9) Light Driver and Osc Module	1	65A27D601
(10) 1 K.C. Amplifier Module	2	65A27D701
(11) Warning Set, Radar Altimeter AN/APQ-107	1	65A27D1

* 3.4.8 Compatibility with Dual Intercommunication Sets - The equipment covered by this specification shall provide two aural warning signal outputs, each to be tied respectively into the pilot's and copilot's intercommunication system. All external signals appearing across the pilot's headset shall be attenuated through the AN/APQ-107 more than 50 db at the copilot's headset and conversely. The output impedance of each AN/APQ-107 aural warning output shall be less than 275 ohms when the AN/APQ-107 is energized and less than 425 ohms when the AN/APQ-107 is de-energized.

* 3.4.9 Spurious Warning Signals - False or spurious warning signals shall not be generated internally in the Radar Altimeter Warning Set, nor shall warning signals be generated by factors external to the Warning Set, except as specified herein.

* 3.4.10 Self Test Provision - A manual "press-to-test" switch shall be mounted on the front surface of the equipment case and provision shall be made for connecting a remotely located test switch. The switch when actuated shall apply the test command input signal of paragraph 3.4.4.6.

3.4.11 Controls - With the exception of the previously mentioned switch, no controls shall be required for the operational use of this equipment. All variable adjustments required for calibration of the Radar Altimeter Warning Set shall be located in such a manner that adjustments can be made without removal of the equipment from the aircraft.

* 3.4.12 Electrical Connections - A single connector, MS 3112E-20-41P, shall be mounted on the unit which shall accommodate all electrical connections essential for the operation of this system. Electrical connections to the pins of this connector shall be as shown on BuWeps Installation Control Drawing 65A27F4.

* 3.4.13 Form Factor and Mounting - The Radar Altimeter Warning Set shall be contained in a single case meeting the applicable requirements of Specification MIL-C-172. The form factor, mounting, mounting provisions, connector, and controls shall conform to BuWeps Installation Control Drawing 65A27F4. Total volume shall be a minimum consistent with good design and shall not exceed 265 cubic inches.

* 3.4.14 Equipment Modifications - Internal modifications to the associated radar altimeter set or other associated equipment shall not be required in order to install or use the warning set. Wiring and other electrical and/or mechanical aircraft modifications necessary for single or dual installation of the Warning Set shall be a minimum consistent with

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good design but shall permit installation of the dual system in separated locations in the aircraft.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of paragraph 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

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4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractor's expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the produc-

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tion equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality, and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of paragraph 4.4.3 are deleted by contractual action.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run-in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

*4.4.3.1 Reliability Procedure - Reliability Procedure I from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be III from Table I of Specification MIL-R-23094. During each cycle the equipment shall be operated at the following time(s):

(1) Power shall be applied continuously during the duty cycle, with all outputs connected to loads duplicating or simulating associated equipment.

(2) At 15 minute intervals, input signals shall be applied in sequence to simulate all warning modes and disabling functions described in the Detail Requirements (paragraph

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3.4). The duration of the simulated inputs shall result in a minimum of 30 seconds operation of the low altitude warning mode, three cycles of operation of the high altitude warning mode, and 30 seconds of operation of each of the other modes.

* 4.4.3.3 Accept-Reject Criteria - Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification Test Plan. Figure 15 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Sampling Test Plan.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated two hours during each cycle.

* 4.4.3.5 Performance Characteristics to be Measured - At the beginning of the heating period, with the equipment at the low temperature limit, and again near the end of the duty cycle, the following characteristics shall be measured:

(1) Functional test of all enabling modes (paragraphs 3.4.3.4 and 4.4.3.2).

(2) Disabling modes (paragraph 3.4.3.5).

(3) Functional test of all controls (paragraph 3.4.11).

(4) Stability of altitude index settings (paragraphs 3.4.1 and 3.4.2).

(5) Electrical aural warning signal (paragraph 3.4.3.1).

(6) Delayed reliability signal (paragraph 3.4.3.3).

(7) Electrical visual warning signal (paragraph 3.4.3.2).

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance Requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - During the period of the tests no preventive measures may be performed upon the equipment except that the calibration adjustments specified in the Detail Requirements (paragraph 3.4) may be readjusted at 500 hour intervals.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipment for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance of special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following, except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests).

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
DC Voltage	27.0 ± 2.0 volts
AC Voltage	115 ± 5 volts (at lowest applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

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(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life tests, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

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6.1 Intended Use - The equipment is intended to provide a low altitude warning system in aircraft equipped with electronic altimeters. The equipment will be used primarily in aircraft engaged in ASW operations or similar low altitude missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 21 to 29 V DC, 105 to 122 V AC, 400 cps, and as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-to-failure greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.6 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

* 6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing - (see paragraph 5.1).

(c) Design Data (See 6.9).

* 6.8 Asterisk - An asterisk "*" preceding a paragraph number denotes paragraphs where changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the asterisk notations and relationship to the last previous issue.

* 6.9 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-24011A(WP) Radar Altimeter Warning Set AN/APQ-107 (*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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